



AIR FORCE INSTITUTE OF TECHNOLOGY
FACULTY OF COMPUTING
DEPARTMENT OF COMPUTER SCIENCE
FIRST SEMESTER EXAMINATION 2024/25
DEGREE EXAMINATION

Course Title: DATA MANAGEMENT II
Course Code: CSC 404
Credit Unit: 3 Units
Instruction: ANSWER QUESTION ONE AND ANY OTHER FOUR (4)
Duration: 3 HOURS
Date: 21 JULY 2025

Question 1. (22 marks)

- a. In terms of data management, write short notes on the following:
- i. Information management (2 marks)
 - ii. Mention the five key areas of information management (2½ marks)
 - iii. Define an Information System (2 marks)
 - iv. Mention any two (2) purposes of information capture, according to ISO 15489, the global standard for records management. (2 marks)
- b. Using a pyramid, list the **types** of information systems and group them according to the **levels** of information systems. (5 marks)
- c. Consider the document type descriptor (DTD) for a book store XML in Appendix A. (4½ marks)
- i. What are the categories of items sold in the book store?
 - ii. Must the book store have those major elements every time?
 - iii. What is the maximum and minimum number of books the bookstore can have
 - iv. What is the maximum and minimum number of magazines the bookstore can have
 - v. Mention the optional and mandatory attributes of a book
 - vi. If a book does not have an edition, will the XML document validate?
 - vii. What is the minimum and maximum number of authors of a magazine?
 - viii. If the year of publication of a magazine is not stated, will the XML document validate?
 - ix. Considering the XML document as a tree, mention one element which is a leaf node.
- d. Rewrite the SQL query below in relational algebra: (2 marks)
- select sID, sName ⚡ Student where GPA > 3.7*
- e. What is Data Independence of a Database Management System? (2 marks)

Question 2. (12 marks)

- a. Mention any six (6) problems of a file processing system. **(3 marks)**
- b. Mention the type of inconsistency that is occurring in each of the following pairs of concurrent statements issued by a client in the figures below: **(3 marks)**

figure b(i)

Update College Set enrollment = enrollment + 1000
Where cName = 'Stanford'

Concurrent with ...

Update College Set enrollment = enrollment + 1500
Where cName = 'Stanford'

figure b(ii)

Update Apply Set major = 'CS' Where sID = 123
Where cName = 'Stanford'

Concurrent with ...

Update Apply Set decision = 'Y' Where sID = 123

figure b(iii)

Update Apply Set decision = 'Y'
Where sID In (Select sID From Student Where GPA > 3.9)

Concurrent with ...

Update Student Set GPA = (1.1) * GPA Where size > 2500

- c. What is a transaction? Explain how it is related to concurrency and system failures. **(4 marks)**
- d. Mention the four (4) properties of transactions. **(2 marks)**

Question 3. (12 marks)

- a. Consider the relation with the following functional dependencies:

Student (SSN, sName, address, HScore, HScity, GPA, priority)

HScore $\rightarrow$ HSname, HScity

GPA $\rightarrow$ priority

SSN $\rightarrow$ sName, address, GPA

- i. Find $\{SSN, HScore\}^+$. **(4 marks)**
- ii. What is the key of the relation? **(1 mark)**
- iii. Mention all the functional dependencies that violate Boyce-Codd Normal Form (BCNF). **(1 mark)**
- iv. Decompose the relation into a schema that is in Boyce Codd Normal Form (BCNF) based on the three functional dependencies. Decompose based on the order below: **(6 marks)**

$$HScore \rightarrow HSname, HScity$$

$$GPA \rightarrow priority$$

$$SSN \rightarrow sName, address, GPA$$

Question 4. (12 marks)

- a. Mention any two (2) of the three commonly used translation schemes used in translating subclass relationships to relations. **(1 mark)**
- b. Consider a scenario that involves *Students* who are either *Foreign students* or *Domestic students*. Also, some of the students are Advanced Placement students also known as *AP students* so, these AP students take *AP courses*.
 - i. Represent this scenario in a UML diagram, showing the subclass relationships, association and association class. **(8 marks)**
Hint:

Each student has a name and a unique matric number; Foreign students have countries they come from; Each domestic student has a state he/she comes from and also has a unique NIN (National Identity Number); An AP course has a course title and a unique course code. The year that an AP student takes a course and the grade he or she gets during the AP course is also important.

(Note! No mark will be awarded if any class is missing)
 - ii. Use the translation scheme defined as “Subclass relations contain superclass key + specialized attribute” to translate the UML in (i) above to relations. **(3 marks)**

Question 5. (12 marks)

- a. List four (4) reasons why we use integrity constraints. **(2 marks)**
- b. Suppose you are using an SQLite database. Create a trigger R1 that will intercept insertions into the *Student* table and will check the GPA. If the GPA of the inserted student is > 3.3 or ≤ 3.6 , that student will be automatically applying to Stanford for a geology major and applying to MIT for a biology major. **(5 marks)**
- c. Consider the relation below, which has the multivalued dependency that: $NIN \twoheadrightarrow hobby$. Derive the remaining tuples based on the multivalued dependency. **(3marks)**

123	piano	CS
123	boxing	EE

- d. There are two types of normal forms. Mention them and also mention the respective properties of a relation that guarantees each of the two normal forms. **(2 marks)**

Question 6. (12 marks)

- a. List the two (2) underlying data structures of an index? **(2 marks)**
- b. Mention the conditions that each of the underlying data structures of an index which you mentioned above can be used. **(2 marks)**
- c. Suppose we have a user that we want to authorize to access i.e., read information in the *Student* relation, but only for students who have applied to AFIT.
- i. Mention how we achieve such an authorization. **(1 mark)**
 - ii. Then show how we can authorize such a user for that specific privilege. **(5 marks)**
 - iii. Then state the required privilege. **(2 mark)**

Question 7. (12 marks)

- a. What is a 'dirty' data item in a database? **(2 marks)**
- b. Mention and describe each of the three levels of data abstraction in a database management system. **(4½ marks)**
- c. In a tabular form, show the comparison between a relational model and XML. The comparison should be based on structure, schema, queries, ordering and implementation. (You can use a word or phrase to describe each of the three models.) **(Total: 2½ marks)**
- d. Use expression trees [in relational algebra] to display the list of GPAs of students applying to Computer Science in Kaduna. **(3 marks)**

Appendix A

<!ELEMENT Bookstore (Book|Magazine)*>

<!ELEMENT Book (Title,Authors,Remark?)>

<!ATTLIST Book ISBN CDATA #REQUIRED

Price CDATA #REQUIRED

Edition CDATA #REQUIRED>

<!ELEMENT Magazine (Title)>

<!ATTLIST Magazine Month CDATA #REQUIRED Year CDATA #REQUIRED>

<!ELEMENT Title (#PCDATA)>

<!ELEMENT Authors (Author+)>

<!ELEMENT Remark (#PCDATA)>

<!ELEMENT Author (First_Name,Last_Name)>

<!ELEMENT First_Name (#PCDATA)>

<!ELEMENT Last_Name (#PCDATA)>

Appendix B

Simple college admission database:

College(cName, state, enrollment)

Student(sID, sName, GPA, sizeHS)

Apply(sID, cName, major, decision)